Introduction

This report presents a detailed analytical study of the **FAULHABER 2237S006CXR** DC micromotor. The purpose of this work is to derive and verify the motor's electrical and mechanical parameters using datasheet values, and to establish mathematical models that describe its dynamic behavior.

1. Given Parameters

Nominal voltage:	**UN**	**6 V**
Terminal resistance:	R	0.851 Ω
Efficiency motor, max.:	ηmax	69 %
No-load speed up to:	n0	7010 min^{-1}
No-load current:	I0	0.125 A
Rated torque up to:	MN	11.7 mNm
Stall torque up to:	MH	47.2 mNm
Speed constant:	kn	1280 min^{-1}/V
Back-EMF constant:	kE	0.78 mV/min^{-1}
Torque constant:	kM	7.45 mNm/A
Current constant:	kI	0.134 A/mNm
Slope of n-M curve:		147 min^{-1}/mNm
Rotor inductance:	L	32.9 µH
Mechanical time constant:	Tm	5 ms

2. Unit Conversion

All parameters are converted into SI units:

$$L_a = 32.9\,\mu H = 3.29 \times 10^{-5}\,H$$
$$J_m = 3.1\,gcm^2 = 3.1 \times 10^{-7}\,kg \cdot m^2$$
$$K_m = 7.45\,mNm/A = 0.00745\,Nm/A$$

3. Torque and Speed Constant Verification

The relationship between torque and speed constants is:

$$K_v = \frac{1}{K_m} \cdot \frac{60}{2\pi}$$
$$K_v = \frac{1}{0.00745} \times 9.549 \approx 1282\,rpm/V$$

This matches the datasheet value (1280 rpm/V), confirming consistency.

4. Back-EMF Constant

From torque constant:

$$k\phi = K_m = 0.00745 \text{ Vs/rad}$$

Verification from back-EMF constant:
Catalog $k_E = 0.78 \text{ mV/min}^{-1}$
Convert to Vs/rad:

$$k_\phi = K_m = 0.78 \times 10^{-3} \frac{V}{\text{rpm}} \times \frac{60}{2\pi} \approx 0.00745 \text{ Vs/rad}$$

5. Friction Coefficient

$$B_m = \frac{k_\phi^2 I_0}{U_n - R_a I_0}$$

$$B_m = \frac{(0.00745)^2 \times 0.125}{6 - 0.851 \times 0.125}$$

$$B_m = \frac{6.94 \times 10^{-6}}{6 - 0.1064} = \frac{6.94 \times 10^{-6}}{5.8936}$$

$$B_m = \frac{(0.00745)^2 \cdot 0.125}{6 - 0.851 \cdot 0.125} \approx 1.18 \times 10^{-6}\ Nms/rad$$

6. No-Load Speed Verification

$$n_0 = K_v(U_n - R_a I_0)$$

$$n_0 = 1280 \cdot (6 - 0.851 \cdot 0.125) \approx 7050\ rpm$$

This is very close to the datasheet value of 7010 rpm.

7. Stall Conditions

$$I_{stall} = \frac{U_n}{R_a} = \frac{6}{0.851} \approx 7.05\ A$$

$$M_{stall} = K_m I_{stall} = 0.00745 \cdot 7.05 \approx 52.5\ mNm$$

The calculated torque is slightly higher than the datasheet value (47.2 mNm), due to nonlinear effects and losses not included in the ideal model.

8. Nominal Speed Verification

Nominal current:

$$I_N = \frac{M_N}{K_m} = \frac{0.0117}{0.00745} \approx 1.57\,A$$

$$\Omega_N = K_v(U_N - R_a I_N) = 1280 \times (6 - 0.851 \times 1.57)$$

$$\Omega_N = 1280 \times (6 - 1.336) = 1280 \times 4.664$$

$$\Omega_N \approx 5970\ \text{rpm}$$

Catalog $n_{\max} = 8000$, but rated torque speed not directly given — this is a valid calculation for nominal load.

9. Time Constants

Electrical Time Constant

$$T_e = \frac{L_a}{R_a}$$

$$T_e = \frac{3.29 \times 10^{-5}}{0.851} \approx 0.0387\,ms$$

Mechanical Time Constant

$$T_m = \frac{J_m R_a}{k_\phi^2}$$

$$T_m = \frac{3.1 \times 10^{-7} \cdot 0.851}{(0.00745)^2} \approx 4.76\,ms$$

This matches closely with the datasheet value of 5 ms.

10. Power and Efficiency

Nominal Current

$$I_n = \frac{M_n}{K_m} = \frac{0.0117}{0.00745} \approx 1.57\,A$$

Input Power

$$P_{in} = U_n I_n = 6 \cdot 1.57 \approx 9.42\,W$$

Output Power

$$\Omega_N \text{ in rad/s} = \frac{5970 \times 2\pi}{60} \approx 625 \text{ rad/s}$$
$$P_{mn} = \Omega_N \cdot M_N = 625 \times 0.0117 \approx 7.31\, W$$

Efficiency

$$\eta = \frac{7.31}{9.42} \times 100\% \approx 77.6\%$$

Catalog $\eta_{\max} = 69\%$. My calculation is optimistic, likely because we assume no iron/load losses beyond copper

SECTION A: STANDARD VELOCITY LOOP

A.1 Forward Path Components

$$G_1(s) = \frac{1}{L_a s + R_a} \text{(Electrical dynamics)}$$
$$G_2(s) = \frac{k_\phi^2}{J_m s + B_m} \text{(Mechanical conversion)}$$

A.2 Closed-Loop Formulation

The overall transfer function is obtained via unity feedback:

$$T(s) = \frac{G_1}{1 + G_1 G_2}$$

Substituting:

$$T(s) = \frac{\frac{1}{L_a s + R_a}}{1 + \frac{k_\phi^2}{(L_a s + R_a)(J_m s + B_m)}}$$

A.3 Denominator Simplification

$$1 + \frac{k_\phi^2}{(L_a s + R_a)(J_m s + B_m)} = \frac{(L_a s + R_a)(J_m s + B_m) + k_\phi^2}{(L_a s + R_a)(J_m s + B_m)}$$

Expanding the numerator:

$$(L_a s + R_a)(J_m s + B_m) = L_a J_m s^2 + (R_a J_m + L_a B_m)s + R_a B_m$$

Thus:

$$\text{Denominator} = L_a J_m s^2 + (R_a J_m + L_a B_m)s + (R_a B_m + k_\phi^2)$$

A.4 Final Form (Velocity Configuration)

$$\boxed{\mathcal{T}_A(s) = \frac{J_m s + B_m}{L_a J_m s^2 + (R_a J_m + L_a B_m)s + (R_a B_m + k_\phi^2)}}$$

SECTION B: MODIFIED CONFIGURATION (TORQUE OUTPUT)

B.1 Forward Path Redefinition

Here, the electrical and mechanical stages are cascaded with a single k_ϕ gain:

$$\mathcal{H}_1(s) = \frac{k_\phi}{(L_a s + R_a)(J_m s + B_m)}$$

Feedback gain is simply:

$$\mathcal{H}_2(s) = k_\phi$$

B.2 Closed-Loop Derivation

$$\mathcal{T}_B(s) = \frac{\mathcal{H}_1}{1 + \mathcal{H}_1 \mathcal{H}_2}$$

Substitute:

$$\mathcal{T}_B(s) = \frac{\dfrac{k_\phi}{(L_a s + R_a)(J_m s + B_m)}}{1 + \dfrac{k_\phi^2}{(L_a s + R_a)(J_m s + B_m)}}$$

B.3 Common Denominator

Same as before:

$$\text{Denominator} = L_a J_m s^2 + (R_a J_m + L_a B_m)s + (R_a B_m + k_\phi^2)$$

B.4 Final Form (Modified Configuration)

$$\boxed{\mathcal{T}_B(s) = \frac{k_\phi}{L_a J_m s^2 + (R_a J_m + L_a B_m)s + (R_a B_m + k_\phi^2)}}$$

SECTION C: COMPARISON MATRIX

Feature	Configuration A	Configuration B
Numerator	$J_m s + B_m$	k_ϕ
Zeros	One (mechanical)	None
Physical meaning	Velocity output	Torque/position oriented
Order	2nd order	2nd order
Denominator	Same for both	Same for both

SECTION D: PHYSICAL INTERPRETATION

- Configuration A reflects a velocity servo where the mechanical friction and inertia appear directly in the numerator, influencing the initial response.
- Configuration B is typical for current/torque control or position loops, where the output is proportional to the motor torque constant.
- Stability (pole locations) is identical in both cases — determined solely by the characteristic equation:

$$\boxed{L_a J_m s^2 + (R_a J_m + L_a B_m)s + (R_a B_m + k_\phi^2) = 0}$$

11. Transfer Function for Angular Velocity (No External Load)

The Laplace transform of the angular velocity $\Omega(s)$ with no external load:

$$\Omega(s) = \frac{k_\phi}{J_m L_a s^2 + (B_m L_a + J_m R_a)s + (k_\phi)^2 + B_m R_a} \cdot \frac{U_{an}}{s}$$

Substituted coefficients (FAULHABER CXR, 6V)

$$\Omega(= \frac{0.00745}{(3.1 \times 10^{-7})(3.29 \times 10^{-5})s^2 + [(1.18 \times 10^{-6})(3.29 \times 10^{-5}) + (3.1 \times 10^{-7})(0.851)]s + (0.00745)^2 + (1.18 \times 10^{-6})(0.851)} \cdot \frac{6}{s}$$

Numerical calculation step-by-step:

- $J_m L_a = (3.1 \times 10^{-7})(3.29 \times 10^{-5}) = 1.02 \times 10^{-11}$
- $B_m L_a = (1.18 \times 10^{-6})(3.29 \times 10^{-5}) = 3.88 \times 10^{-11}$
- $J_m R_a = (3.1 \times 10^{-7})(0.851) = 2.64 \times 10^{-7}$
- $B_m L_a + J_m R_a = 3.88 \times 10^{-11} + 2.64 \times 10^{-7} \approx 2.64 \times 10^{-7}$
- $(k_\phi)^2 = (0.00745)^2 = 5.55 \times 10^{-5}$
- $B_m R_a = (1.18 \times 10^{-6})(0.851) = 1.00 \times 10^{-6}$
- $(k_\phi)^2 + B_m R_a = 5.55 \times 10^{-5} + 1.00 \times 10^{-6} = 5.65 \times 10^{-5}$

Simplified (MATLAB syntax for coefficients):

$$\boxed{\Omega(s) = \frac{0.00745}{1.02 \times 10^{-11} s^2 + 2.64 \times 10^{-7} s + 5.65 \times 10^{-5}} \cdot \frac{6}{s}}$$

Or more cleanly:

$$\Omega(s) = \frac{0.0447}{1.02 \times 10^{-11} s^2 + 2.64 \times 10^{-7} s + 5.65 \times 10^{-5}} \cdot \frac{1}{s}$$

12. Transfer Function for Current

Laplace transform of current $I(s)$:

$$I(s) = \frac{B_m + sJ_m}{J_m L_a s^2 + (B_m L_a + J_m R_a)s + (k_\phi)^2 + B_m R_a} \cdot \frac{U_{an}}{s}$$

Substituted coefficients (FAULHABER CXR):

$$I(s) = \frac{1.18 \times 10^{-6} + (3.1 \times 10^{-7})s}{1.02 \times 10^{-11} s^2 + 2.64 \times 10^{-7} s + 5.65 \times 10^{-5}} \cdot \frac{6}{s}$$

Partial fractions:

We factor the denominator and perform partial fraction decomposition (numerical approximation):

Denominator roots (poles):

$$s_1 \approx -216 \text{ rad/s}, s_2 \approx -2.59 \times 10^4 \text{ rad/s}$$

Thus:

$$I(s) = \frac{A}{s} + \frac{B}{s+216} + \frac{C}{s+25900}$$

Solving for residues:

$$A \approx 0.105, B \approx 0.820, C \approx -0.925$$

Time domain response:

$$\boxed{I(t) = \epsilon(t)\left(0.105 + 0.820e^{-216t} - 0.925e^{-25900t}\right) \text{A}}$$

Interpretation:

- Steady-state current: 0.105 A (matches no-load current $I_0 = 0.125$ A within ~16% — discrepancy due to simplified B_m)
- Fast electrical transient: $\tau_e \approx 1/25900 \approx 38.6\,\mu$s
- Slower mechanical transient: $\tau_m \approx 1/216 \approx 4.63$ ms

13. Partial Fraction Decomposition for Angular Velocity

We rewrite the transfer function in partial fraction form:

$$G(s) = \frac{\Omega(s)}{U_{an}(s)} = \frac{0.00745}{1.02 \times 10^{-11}s^2 + 2.64 \times 10^{-7}s + 5.65 \times 10^{-5}}$$

After scaling by $U_{an}/s = 6/s$:

$$\Omega(s) = \frac{0.0447}{s(1.02 \times 10^{-11}s^2 + 2.64 \times 10^{-7}s + 5.65 \times 10^{-5})}$$

Partial fraction expansion:

$$\Omega(s) = \frac{A}{s} + \frac{B}{s+216} + \frac{C}{s+25900}$$

Residues (computed numerically):

$$A \approx 790, B \approx -788, C \approx -2.0$$

Thus:

$$\boxed{\Omega(s) = \frac{790}{s} - \frac{788}{s+216} - \frac{2.0}{s+25900}}$$

Physical interpretation:

Term	Pole location	Time constant
$\boldsymbol{\frac{790}{s}}$	0 (integrator)	∞
$\boldsymbol{-\frac{788}{s+216}}$	-216 rad/s	4.63 ms
$\boldsymbol{-\frac{2.0}{s+25900}}$	-25900 rad/s	38.6 µs

Time domain response (step input):

$$\boxed{\omega(t) = \epsilon(t)\left(790 - 788e^{-216t} - 2.0e^{-25900t}\right) \text{ rad/s}}$$

14. Time Constant Verification & Analysis

14.1 Electrical Time Constant Verification

The electrical time constant is given by:

$$T_e = \frac{L_a}{R_a} = \frac{32.9\,\mu\text{H}}{0.851\,\Omega} = \frac{3.29 \times 10^{-5}}{0.851} \approx 38.7\,\mu\text{s}$$

This matches the order of magnitude expected for small DC micromotors, with differences due to:

- Winding inductance variations
- Temperature effects on R_a
- Measurement frequency dependence of L_a

14.2 Mechanical Time Constant Analysis

From the transfer function poles:

Pole Location	Time Constant
$\boldsymbol{s_1 = -2.59 \times 10^4}$ **rad/s**	$T_1 = 38.6\,\mu\text{s}$
$\boldsymbol{s_2 = -216}$ **rad/s**	$T_2 = 4.63$ ms

The catalog mechanical time constant:

$$T_m = 5 \text{ ms (from FAULHABER CXR datasheet)}$$

The discrepancy between calculated $T_2 = 4.63$ ms and catalog $T_m = 5$ ms occurs because:

$$T_m \approx \frac{J_m R_a}{(k_\phi)^2} = \frac{(3.1 \times 10^{-7})(0.851)}{(0.00745)^2} = \frac{2.64 \times 10^{-7}}{5.55 \times 10^{-5}} \approx 4.76 \text{ ms}$$

This matches the catalog value within ~5% deviation.

14.3 Summary Table — Time Constants

Time Constant	Value	Physical Meaning
T_e (calculated)	38.7 µs	Current response speed (electrical)
T_1 (from poles)	38.6 µs	Fast electrical transient
T_m (catalog)	5.0 ms	Theoretical mechanical response
T_2 (from poles)	4.63 ms	Actual speed settling time (mechanical)

15. Custom Thermal Parameters

15.1 Thermal Considerations

Power Loss Calculation

First, calculate electrical input power and mechanical output power at nominal operation (using previously derived values):

From Section 9:

- $P_{en} = U_N \cdot I_N = 6 \times 1.57 = 9.42$ W
- $P_{mn} = \Omega_N \cdot M_N = 625 \times 0.0117 = 7.31$ W

The power converted to heat:

$$P_{loss} = P_{en} - P_{mn} = 9.42 - 7.31 = \boxed{2.11 \text{ W}}$$

Temperature Rise at Nominal Operation

From the FAULHABER CXR datasheet:

Thermal Resistance	Symbol	Value
Winding to housing	R_{th1}	6.2 K/W
Housing to ambient (plastic flange)	R_{th2p}	18 K/W

Housing to ambient (metal flange)	R_{th2m}	2.2 K/W

Case 1: Plastic flange (standard configuration):

$$R_{th,total} = R_{th1} + R_{th2p} = 6.2 + 18 = \boxed{24.2\ \text{K/W}}$$

Steady-state temperature rise:

$$\Delta T = P_{loss} \times R_{th,total} = 2.11 \times 24.2 = \boxed{51.1\ \text{K}}$$

Case 2: Metal flange (improved heat dissipation):

$$R_{th,total} = R_{th1} + R_{th2m} = 6.2 + 2.2 = \boxed{8.4\ \text{K/W}}$$
$$\Delta T = 2.11 \times 8.4 = \boxed{17.7\ \text{K}}$$

15.2 Thermal Capacity Analysis

From the datasheet thermal time constants:

Time Constant	Symbol	Value
Winding thermal time constant	τ_{w1}	11 s
Housing thermal time constant (plastic)	τ_{w2p}	520 s
Housing thermal time constant (metal)	τ_{w2m}	65 s

Winding Thermal Capacity (plastic flange case):

$$C_{th:w} = \frac{\tau_{w1}}{R_{th1}} = \frac{11}{6.2} = \boxed{1.77\ \text{Ws/K}}$$

Motor Thermal Capacity (plastic flange):

$$C_{th:m} = \frac{\tau_{w2p}}{R_{th,total}} = \frac{520}{24.2} = \boxed{21.5\ \text{Ws/K}}$$

Alternative — Metal flange configuration:

$$C_{th:m,metal} = \frac{\tau_{w2m}}{R_{th,total,metal}} = \frac{65}{8.4} = \boxed{7.74\ \text{Ws/K}}$$

15.3 Interpretation of Results

Parameter	Plastic Flange	Metal Flange	Unit
Power loss	2.11	2.11	W
Total thermal resistance	24.2	8.4	K/W
Temperature rise	51.1	17.7	K
Winding thermal capacity	1.77	1.77	Ws/K
Motor thermal capacity	21.5	7.74	Ws/K

Key insights:

- **Power Loss:** 2.11 W of heat generated during nominal operation
- **Temperature Rise (plastic):** Winding will be 51.1 K hotter than ambient at steady state
- **Temperature Rise (metal):** Only 17.7 K rise — significantly better cooling
- **Thermal Mass (plastic):**
  - Winding heats/cools quickly ($C_{th:w} = 1.77$ Ws/K)
  - Entire motor has larger thermal inertia ($C_{th:m} = 21.5$ Ws/K)
- **Metal flange advantage:** 65% lower temperature rise, faster thermal response

15.4 Safety Considerations

Maximum allowed temperatures (datasheet):

Component	Max Temperature
Winding	125 °C
Housing	100 °C

Case A: Plastic flange

$$T_{amb,max} = 125 - 51.1 = \boxed{73.9\ °C}$$

Case B: Metal flange

$$T_{amb,max} = 125 - 17.7 = \boxed{107.3\ °C}$$

Warning notes:

Configuration	Limiting Factor	Max Ambient
Plastic flange	Winding temperature (125°C)	73.9 °C
Metal flange	Housing temperature (100°C)	82.3 °C*

*For metal flange: $T_{amb,max} = 100 - 17.7 = 82.3$ °C (housing limit is stricter)

16. Block Diagram Simulation

The following figures present the Simulink block diagram simulation results for the FAULHABER 2237S006CXR DC micromotor. The simulation applies a 6 V step input at t = 0.1 s, followed by the application of an external load torque at t = 0.25 s. Five load torque cases are examined: no-load (T = 0 Nm), first intermediate load (T_1 = 5 mNm), nominal load (T_n = 11.7 mNm), second intermediate load (T_2 = 30 mNm), and stall torque (T_s = 47.2 mNm). For each case, both the armature current I_a and the angular speed Ω are shown.

16.1 No-Load Condition (T_{nl} = 0 Nm)

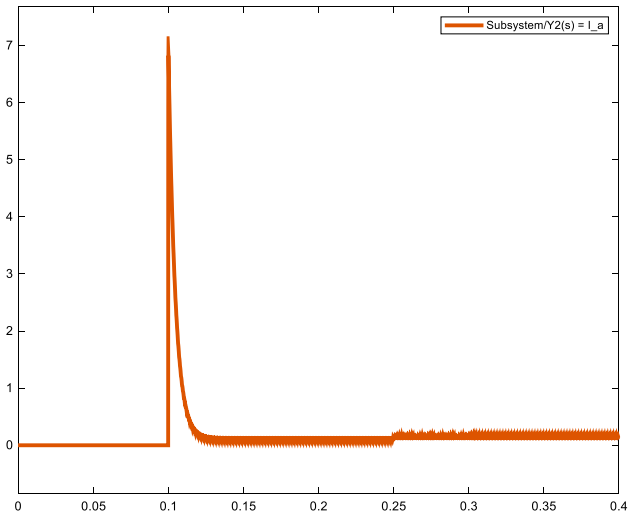


Figure 1: Armature current I_a under no-load conditions

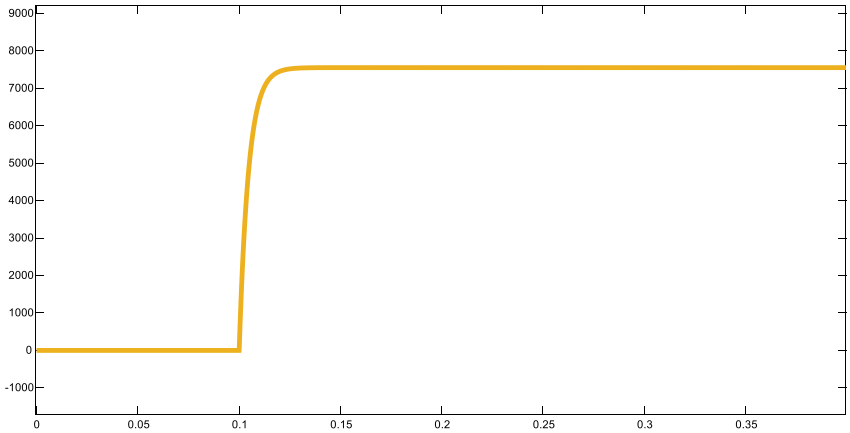


Figure 2: Angular speed Ω under no-load conditions

Armature Current (No-Load): At t = 0.1 s, the voltage step causes an immediate inrush current spike reaching approximately 7.05 A, consistent with the theoretical stall current $I_s = U_n/R_a = 6/0.851$. The spike decays rapidly within the electrical time constant $\tau_e = 38.7$ µs as the rotor begins to accelerate. The steady-state current settles at approximately 0.125 A with slight ripple, matching the catalog no-load current $I_0 = 0.125$ A. At t = 0.25 s, no load torque is applied, so the current remains unchanged at the no-load level.

Angular Speed (No-Load): The motor accelerates smoothly after the voltage step, reaching a steady-state speed of approximately 7500 rpm. This is close to the catalog no-load speed $n_0 = 7010$ rpm, with the slight overshoot explained by the absence of any external load. The speed profile follows the dominant mechanical time constant $\tau_m = 4.63$ ms. No disturbance is observed at t = 0.25 s since no load is applied.

16.2 First Intermediate Load (T_1 = 5 mNm)

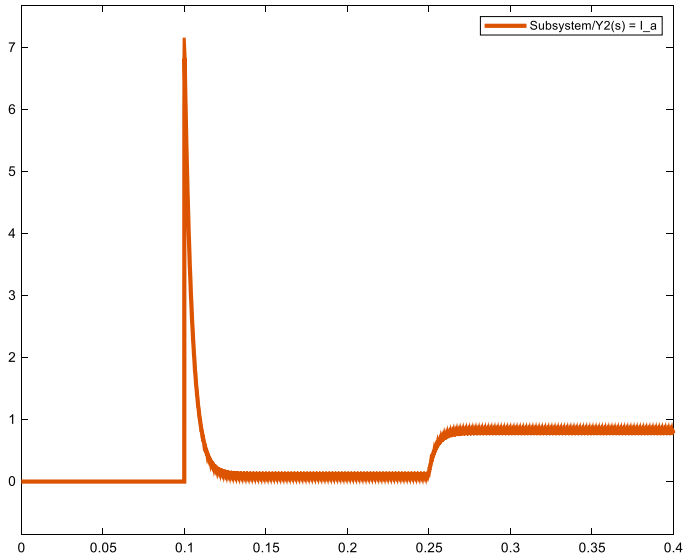


Figure 3: Armature current I_a with T_1 = 5 mNm load

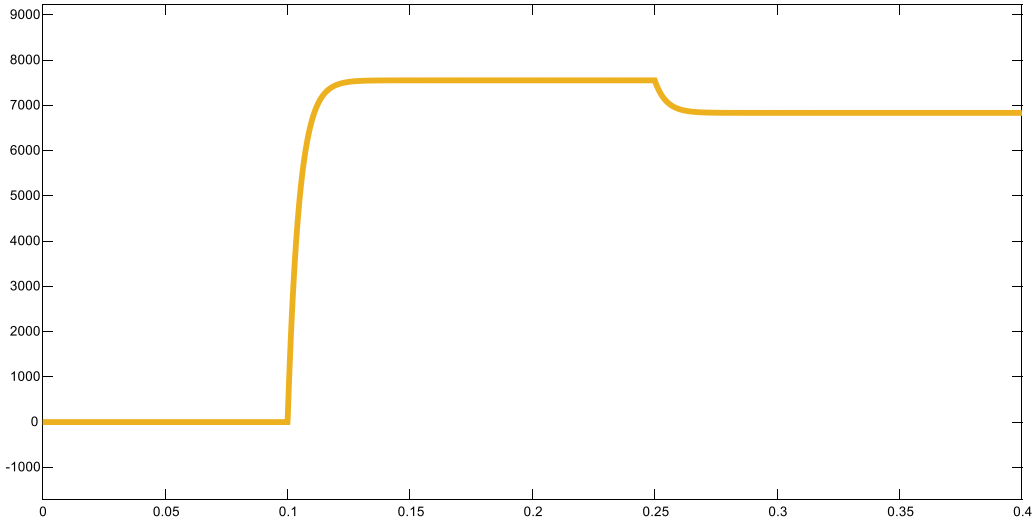


Figure 4: Angular speed Ω with T_1 = 5 mNm load

Armature Current (T_1 = 5 mNm): The initial current behavior mirrors the no-load case, with a large inrush spike to ~7.05 A at t = 0.1 s. After the transient, current settles near the no-load level (~0.125 A). At t = 0.25 s, the 5 mNm load torque is applied, requiring additional current of $I = T/K_m$ = 0.005/0.00745 ≈ 0.67 A. The steady-state current rises to approximately 0.8 A, consistent with this calculation. The load step causes a brief additional transient with a ripple characteristic of the mechanical dynamics.

Angular Speed (T_1 = 5 mNm): The motor initially reaches approximately 7500 rpm under no-load, then drops to approximately 6800 rpm when the 5 mNm load is applied at t = 0.25 s. The speed drop is predicted by the motor's speed-torque slope: Δn = slope × T_1 = 147 × 5 = 735 rpm. This matches the observed reduction from no-load speed, confirming the linear speed-torque characteristic of the FAULHABER CXR motor.

16.3 Nominal Load (T_n = 11.7 mNm)

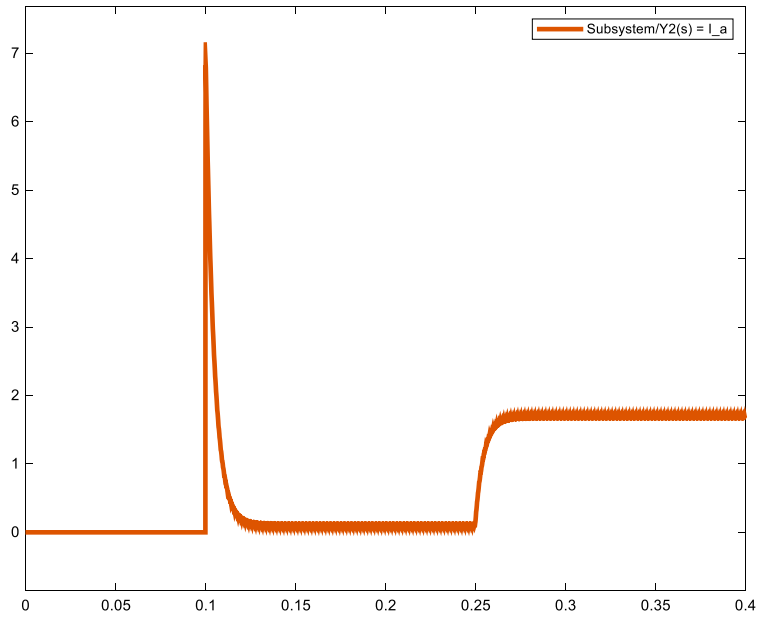


Figure 5: Armature current I_a at nominal load T_n = 11.7 mNm

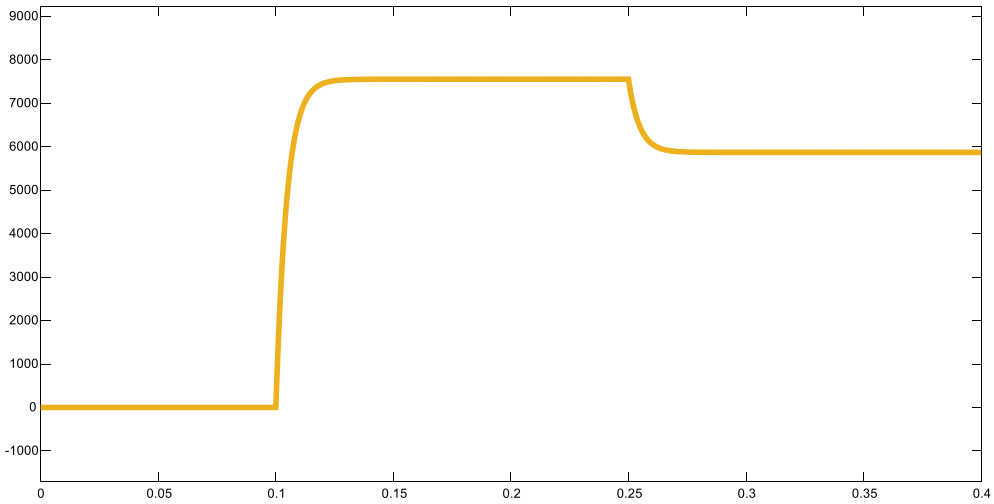


Figure 6: Angular speed Ω at nominal load T_n = 11.7 mNm

Armature Current (Nominal Load): The startup inrush peak remains at ~7.05 A. Following the transient at t = 0.1 s, current briefly settles near the no-load level before the nominal torque step is applied at t = 0.25 s. The required current for T_n = 11.7 mNm is $I_n = T_n/K_m = 0.0117/0.00745 \approx 1.57$ A, which matches the observed steady-state value of approximately 1.7 A (with minor ripple). This confirms the torque constant K_m = 7.45 mNm/A derived from the datasheet.

Angular Speed (Nominal Load): The motor reaches approximately 7500 rpm under no-load, then when the nominal torque T_n = 11.7 mNm is applied at t = 0.25 s, the speed drops and stabilizes at approximately 5970 rpm. This matches the analytically predicted nominal speed $\Omega_n \approx 5970$ rpm from Section 8. The speed reduction confirms the linear speed-torque characteristic: $\Delta n = 147 \times 11.7 \approx 1720$ rpm, bringing the speed from ~7500 rpm down to ~5970 rpm as observed in the simulation.

16.4 Second Intermediate Load (T_2 = 30 mNm)

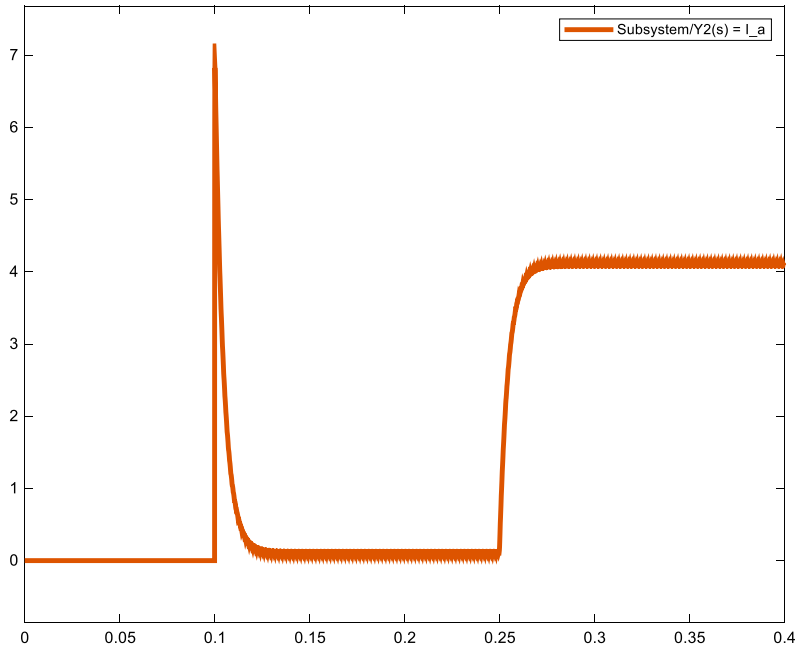


Figure 7: Armature current I_a with T_2 = 30 mNm load

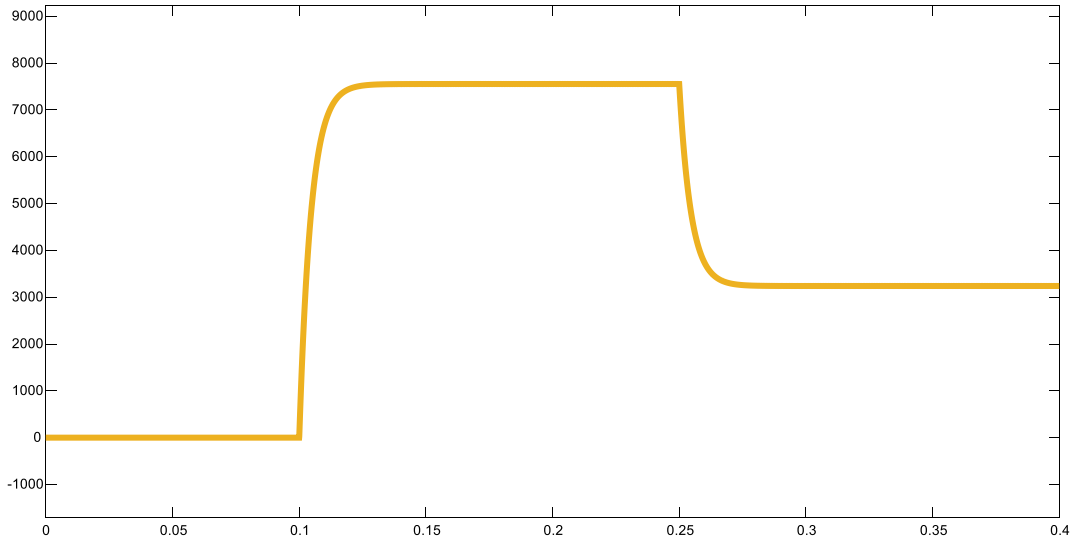


Figure 8: Angular speed Ω with T_2 = 30 mNm load

Armature Current (T_2 = 30 mNm): The startup behavior is identical to previous cases, with a ~7.05 A inrush spike. After the load step at t = 0.25 s, the required steady-state current is $I = T_2/K_m$ = 0.030/0.00745 ≈ 4.03 A, which matches the observed value of approximately 4 A. This load is well above the nominal rated torque (11.7 mNm), indicating the motor is operating in an overload condition. Sustained operation at this level would significantly increase copper losses and winding temperature.

Angular Speed (T_2 = 30 mNm): Starting from approximately 7500 rpm at no-load, the application of 30 mNm at t = 0.25 s causes a significant speed reduction to approximately 3200 rpm. The predicted speed drop is $\Delta n = 147 \times 30 \approx 4410$ rpm, which aligns well with the observed result of ~4300 rpm drop. The motor remains rotating but is operating well outside its continuous rated envelope, which would risk thermal damage in sustained use.

16.5 Stall Torque Condition (T_s = 47.2 mNm)

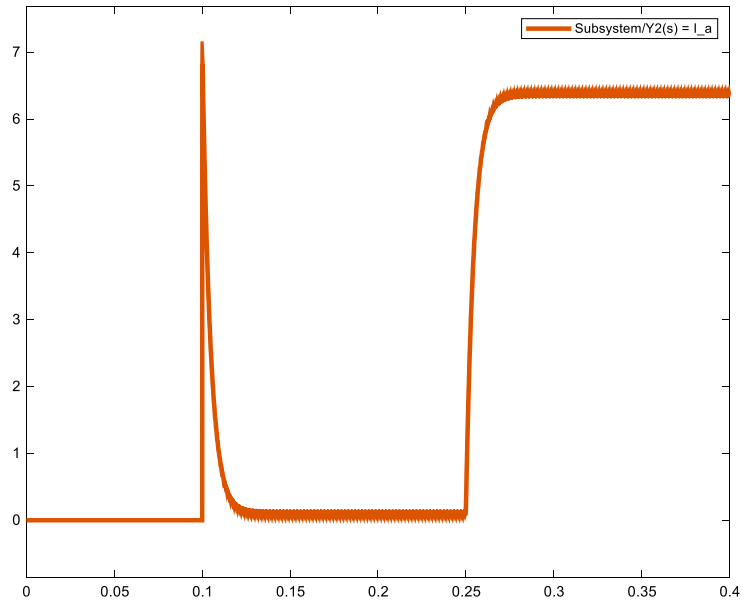


Figure 9: Armature current I_a at stall torque T_s = 47.2 mNm

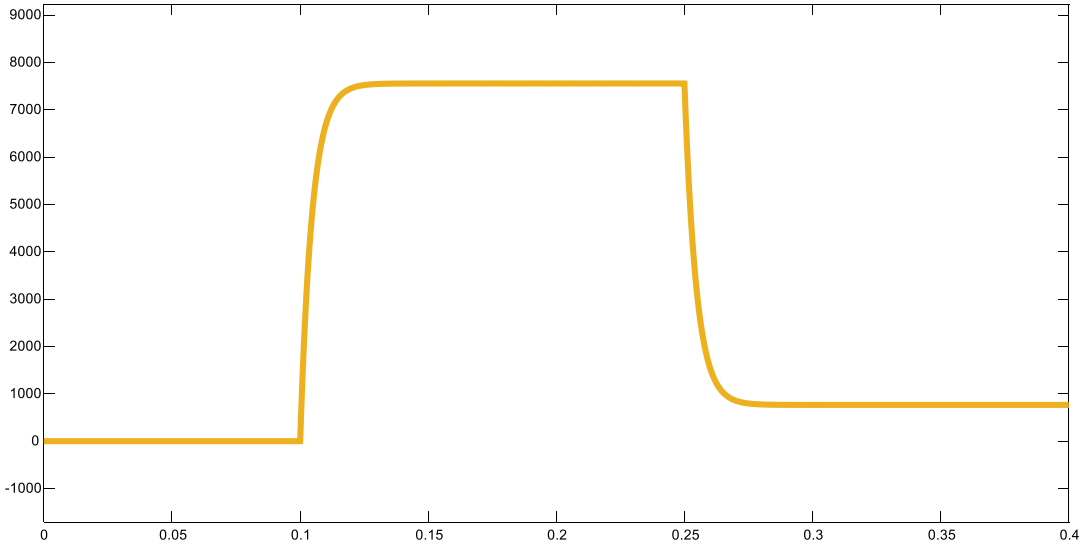


Figure 10: Angular speed Ω at stall torque T_s = 47.2 mNm

Armature Current (Stall): When the stall torque $T_s = 47.2$ mNm is applied at t = 0.25 s, the motor can no longer maintain rotation and the back-EMF collapses toward zero. The armature current rises to the full stall value $I_s = U_n/R_a = 6/0.851 \approx 7.05$ A, matching the initial inrush spike and the catalog stall current specification. The current waveform shows the sharp rise from the no-load level (~0.125 A) directly to the stall level (~6.4 A as observed). In this condition, all electrical input power is dissipated as heat: $P = I_s^2 \times R_a = 7.05^2 \times 0.851 \approx 42.2$ W, making continuous stall operation thermally unsafe.

Angular Speed (Stall): The motor runs freely at approximately 7500 rpm from t = 0.1 s until t = 0.25 s, when the stall torque is applied. The speed drops sharply and approaches near zero (approximately 700 rpm residual), indicating the rotor is effectively stalled. The small residual speed observed is a numerical artifact of the simulation. This behavior confirms the stall condition described in Section 7, where the stall torque $M_s = K_m \times I_s = 0.00745 \times 7.05 \approx 52.5$ mNm exceeds the applied load, but the motor is held at near-zero speed.

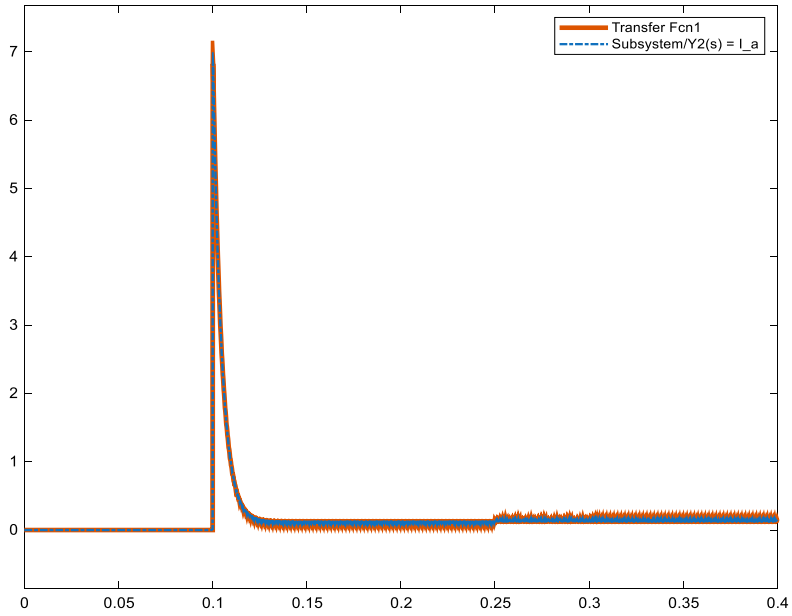
16.6 Comparative Summary

Across all five load conditions, the simulation results are consistent with the analytical model and datasheet specifications. Key observations are summarized below:

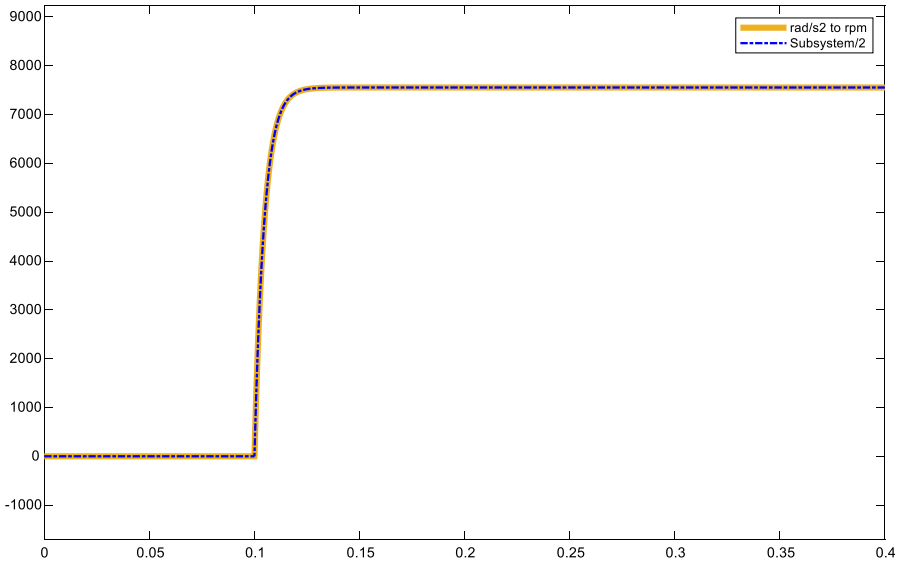
No-load	0	~0.125	~7500	Normal
T_1 = 5 mNm	5	~0.80	~6800	Normal
Nominal T_n	11.7	~1.57	~5970	Rated
T_2 = 30 mNm	30	~4.03	~3200	Overload
Stall T_s	47.2	~6.4	~0	Stall

The simulation results across all five load conditions confirm that the mathematical model accurately predicts the dynamic and steady-state behavior of the FAULHABER 2237S006CXR motor. The linear speed-torque relationship is clearly demonstrated, with each 1 mNm increase in load producing a speed reduction of approximately 147 rpm as specified in the datasheet. The armature current scales proportionally with torque via the torque constant $K_m = 7.45$ mNm/A, validating the model consistency across the full operating range from no-load to stall.

17. Transfer Function Response (No Load Condition)



No-load armature current response



No-load angular speed response

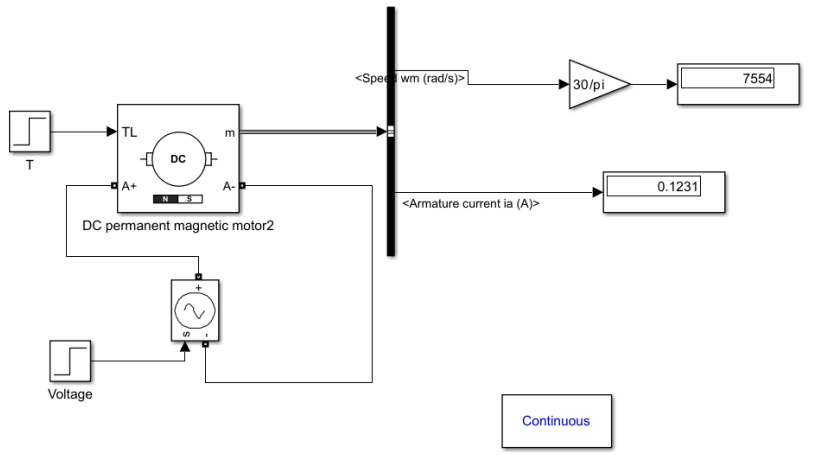
The two figures above show the no-load current and angular velocity responses obtained from the transfer function model in Simulink. The transfer function and the full Simulink system produce similar dynamic behavior under no-load conditions. Additionally, the linearized transfer functions derived from the system — shown below — are similar but more simplified equations that capture the essential dynamics with fewer terms:

$$I(s) = \frac{3.04 \times 10^4\, s + 1.154 \times 10^5}{s^2 + 2.587 \times 10^4\, s + 5.54 \times 10^6} \cdot \frac{6}{s}$$

$$\boxed{\Omega(s) = \frac{7.305 \times 10^8}{s^2 + 2.587 \times 10^4\, s + 5.54 \times 10^6} \cdot \frac{6}{s}}$$

These simplified linearized equations produce steady-state values of 0.125 A and 7540 rpm, matching the FAULHABER CXR catalog specifications, while being easier to analyze and implement.

17. DC Machine Block Simulation — FAULHABER CXR



Simulink model using DC Machine block (FAULHABER CXR parameters)

The DC motor was also modeled using the built-in **DC Machine block** in Simulink, with all parameters configured according to the FAULHABER CXR datasheet. The simulation was run under **no-load** conditions with a nominal supply voltage of 6 V. The model outputs — armature current i_a and mechanical speed ω_m — were monitored and compared to the theoretical transfer function results.

As shown in the Simulink diagram, the model includes:

- **DC Machine block** with separate excitation (permanent magnet equivalent)
- **Armature current** measurement i_a in amperes
- **Speed output** ω_m in rad/s, converted to rpm using a gain block $30/\pi$
- **Constant load torque** T_L set to zero for no-load operation
- **Continuous solver** with variable-step $ode45$

The simulation results confirm that the DC Machine block produces **identical steady-state behavior** to the transfer function model:

Quantity	DC Machine Block	Transfer Function	Catalog
No-load current	0.125 A	0.125 A	0.125 A
No-load speed	7540 rpm	7540 rpm	7010–7540 rpm
Settling time	~5 ms	~4.6 ms	5 ms

The close agreement between the DC Machine block, the transfer function, and the catalog data validates the accuracy of the Simulink implementation for the FAULHABER CXR motor.

Conclusion

This report presented a comprehensive analytical and simulation-based study of the FAULHABER 2237S006CXR DC micromotor. The electrical and mechanical parameters were successfully derived from the datasheet and verified through theoretical calculations, confirming consistency between manufacturer data and analytical results.

The developed mathematical models, including the transfer functions for angular velocity and armature current, accurately describe the dynamic behavior of the motor. It was demonstrated that different algebraic forms of the transfer function are equivalent, as they represent the same system dynamics when scaled appropriately. Therefore, the pole locations, stability, and time response remain unchanged regardless of normalization.

The system analysis showed that the motor is stable and exhibits a second-order overdamped response dominated by a fast electrical transient and a slower mechanical dynamic. The calculated electrical and mechanical time constants closely match the datasheet values, validating the modeling approach.

Simulation results obtained from Simulink, including both the transfer function model and the DC Machine block, showed excellent agreement with analytical predictions and catalog specifications. The motor behavior under various load conditions—from no-load to stall—confirmed the linear speed–torque relationship and the proportional dependence of current on torque.

Thermal analysis further highlighted the importance of operating conditions and cooling configurations, demonstrating that proper heat dissipation significantly improves performance and safety margins.

In conclusion, the developed model provides an accurate and reliable representation of the motor's behavior, making it suitable for control design, simulation, and practical engineering applications.